

Dr. Satyvir Singh

Multiphysics Simulation Engineer | CFD & Thermal Analysis

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Location: Weinstr. 74A, 52477 Alsdorf, Germany

Nationality: Indian | **Work Authorization:** German PR (Niederlassungserlaubnis)

Professional Summary

Multiphysics simulation engineer with 10+ years of experience in CFD, thermal and heat-transfer analysis for complex fluid–structure systems, from continuum to rarefied regimes. Expert in high-fidelity simulations of compressible and non-equilibrium flows, with a strong background in setting up, automating, and validating multiphysics workflows (CFD + heat transfer, aero-thermal loads). Proficient in Python, C++, Fortran, and modern CAE tools (OpenFOAM, ANSYS Fluent, Athena++, Trixi.jl), with a track record of building and optimizing custom solvers and parameter studies to support engineering design decisions. Experienced collaborating with cross-functional and international teams, and presenting complex simulation results to non-CFD stakeholders.

Core Technical Skills & Multiphysics Experience

- **Multiphysics:** CFD–thermal coupling, heat transfer, aero-thermal loading, continuum–rarefied interactions, fluid-mechanics aspects of fluid–structure interaction.
- **Programming & HPC:** C++, Fortran, Python, Julia, Bash, Git, Linux, MPI, HPC/SLURM, Docker; Python scripting for parametric studies and automated simulation workflows.
- **Numerical Methods:** Discontinuous Galerkin (DG), finite volume (FVM), finite element (FEM), explicit/implicit time integration, shock-capturing schemes, multiscale and rarefied flow models.
- **CFD & Physics:** Compressible flows, shock waves, hydrodynamic instabilities (Richtmyer–Meshkov, Rayleigh–Taylor), continuum–rarefied regimes, interface-driven and multiphase flows, non-equilibrium gas dynamics.
- **Tools & Ecosystem:** ANSYS Fluent, OpenFOAM, Athena++, Trixi.jl, Gmsh, Pointwise, ParaView, Tecplot, HDF5/VTK; experienced in CAD/mesh preparation and automated pre/post-processing.
- **Scientific ML:** PyTorch, TensorFlow; physics-informed neural networks (PINNs), neural operators (FNO, DeepONet); data-driven model closure for non-equilibrium flow simulations.

Professional Experience

Scientific Staff / Postdoctoral Research Scientist

Sept 2022 – Present

Institute for Applied and Computational Mathematics, RWTH Aachen University, Germany

- Develop and implement computational models for non-equilibrium and continuum–rarefied gas flows in in-house CFD frameworks, focusing on robust discretization and stability of moment-equation-based solvers.
- Use high-fidelity CFD and moment-equation solvers to predict aero-thermal loads and rarefied-flow effects around complex aerodynamic geometries, supporting design-oriented flow analysis.
- Integrate ML-based moment-closure extraction (Python/PyTorch) into Grad-type models to improve predictive capability of non-equilibrium gas-flow simulations.
- Contribute to numerical method development for continuum–rarefied transition regimes (Grad-13/17, R13), including DG/FEM formulations and verification of solver implementations.
- Collaborate with teams in Europe and Asia on solver development and multiphysics flow analysis.

Postdoctoral Research Fellow – High-Order CFD & Kinetic Theory

Aug 2018 – Jul 2022

School of Physical and Mathematical Sciences, Nanyang Technological University, Singapore

- Built 2D/3D modal DG solvers for the Boltzmann transport equation under electronic and gas-dynamic non-equilibrium, including constitutive models for diatomic and polyatomic gases relevant to high-speed and microscale applications.
- Developed algorithms for shock-induced flow instabilities in compressible configurations; performed extensive parameter studies over Mach number, Atwood number, and geometry for design-space exploration.

- Simulated coupled effects of shocks, thermal non-equilibrium, and material properties on instability growth, providing insight into aero-thermal loading in high-speed configurations.

Senior Researcher – Rarefied and Aerospace Flows

Mar 2018 – Jul 2018

Research Center for Aircraft Core Technology, Gyeongsang National University, South Korea

- Improved computational efficiency and robustness of high-fidelity rarefied-flow simulations used in aerospace certification workflows.
- Implemented and tested boundary-condition models (Langmuir, Maxwell–Smoluchowski) for rarefied gas dynamics within existing solvers, ensuring numerical stability across Knudsen regimes.
- Performed simulations of rarefied hypersonic flows to assess aero-thermal environments for re-entry and high-altitude vehicles.

PhD Researcher – CFD Solver Development

Mar 2013 – Feb 2018

School of Mechanical and Aerospace Engineering, Gyeongsang National University, South Korea

- Designed and implemented a 3D mixed-modal DG solver for second-order Boltzmann–Curtiss-based hydrodynamic models of diatomic and polyatomic gases, targeting continuum–rarefied gas flows.

Selected Solver & Simulation Projects

3D Modal Discontinuous Galerkin Solver for Boltzmann–Curtiss Models – PhD Project

- Developed a scalable 3D DG solver including basis functions, numerical fluxes, slope limiting, and MPI-based domain decomposition; applied to continuum–rarefied gas flows and validated against experimental benchmarks.

Shock-Driven Instability Simulation Suite (Richtmyer–Meshkov Flows) – NTU Singapore & RWTH Aachen

- Performed 35+ high-resolution simulations of shock-accelerated bubbles and interfaces (square/rectangular/cylindrical/elliptical shapes) to analyze instability growth and vorticity production.
- Implemented robust shock-capturing strategies in high-order schemes and used the suite as a verification/validation testbed.

Education

Ph.D. in Aerospace Engineering (Computational Fluid Dynamics)

Mar 2013 – Feb 2018

Gyeongsang National University, South Korea

Thesis: Development of a 3D discontinuous Galerkin method for second-order Boltzmann–Curtiss-based hydrodynamic models of diatomic and polyatomic gases.

M.Tech. in Industrial Mathematics & Scientific Computing

Aug 2009 – Jun 2011

Indian Institute of Technology (IIT) Madras, India

Thesis: A complete heat-line analysis on mixed convection within a square cavity: Effect of aspect ratio and thermal boundary conditions.

Publications & Impact (Abbreviated)

- 60+ peer-reviewed papers in Physics of Fluids, Physical Review Fluids, Journal of Computational Physics, Computers & Fluids, Physica D., Applied and Computational Mathematics
- Recent works include DG solvers for Grad-moment equations, shock-accelerated bubble dynamics, non-equilibrium effects in diatomic/polyatomic gases, and magnetohydrodynamic stabilization of instabilities.
- Full publication list: satyvir-singh.github.io / Google Scholar.

Awards & Scholarships

- Brain Korea 21+ Scholarship for PhD, Gyeongsang National University (2013–2016).
- GATE-MHRD Scholarship for M.Tech., IIT Madras (2009–2011).
- Qualified CSIR-JRF NET in Mathematical Sciences, All-India Rank 38 (2011).

Languages

English (professional), German (A1–A2 progressing), Hindi (native).